

¹H NMR Spectroscopic Study of the Solution Structure of a Conjugated Polymer

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The structure of poly[2-methoxy-5-(2'-ethylhexyloxy)-1,4-phenylenevinylene] (MEH-PPV) solutions with a number of common deuterated organic solvents was studied by means of ¹H NMR spectroscopy. The resonance bands of the polymer backbone protons appear at much lower frequencies in CDCl₃ compared to their positions in aromatic solvents. This has been attributed to the π - π self-stacking interaction of the polymer segments in the aliphatic solvent, which is replaced by a polymer-solvent π - π interaction in the aromatic solvents. In mixed solvent systems with an aliphatic solvent component and an aromatic one, the intensities of the polymer resonances pass through a maximum at intermediate solvent compositions. This has been attributed to the minimum aggregation tendency of the polymer at these solvent compositions. The results suggest that a mixed solvent system constituting an aliphatic component and an aromatic component might be a choice for casting films of hairy-rod type conjugated polymers if minimization of aggregate domains in the films is desired.

Keywords: Conjugated polymer; MEH-PPV; NMR; Solution structure; Polymer-solvent interaction; Pi-pi-stacking; Aggregation.

INTRODUCTION

Rapid rotational Brownian diffusion of molecules in liquid samples largely averages out dipolar and other anisotropic magnetic interactions local to NMR-active nuclei in the molecule. This facilitates motional narrowing of NMR lines due to them. For example, a nucleus in a molecule having diamagnetic anisotropy feels different extents of the applied magnetic field of the NMR at different orientations relative to the field direction, known as chemical shift anisotropy. Due to rapid molecular tumbling we observe only one chemical shift for the nucleus, an average value of all the orientations. Chemical shifts are also sensitive to small variations in the local magnetic field arising from different chemical environments in rotational (conformational) isomers, which are normally averaged in the liquid state due to rapid rotation of the part of the molecule on the NMR timescale. However, if the rotation is slow relative to the latter (lifetime long enough on the NMR timescale), we may observe different chemical shifts due to different rotational isomers.

The Brownian motion of molecules decreases with increasing size. As a consequence, the linewidth (at half height) of an NMR resonance due to a nucleus increases with the increasing size of the molecule to which it is attached. For large rigid molecules the signal would be so broad that it would become indistinguishable from the baseline. That we still can obtain high resolution NMR spectra from polymer molecules in the solution state is not due their overall reorientation; rather it is the local segmental rotation of the macromolecule that gives rise to the high resolution spectra. This allows us to determine microstructural features and dynamics of macromolecules.^{1,2} The intensity of an NMR signal is proportional to the number of nuclei giving rise to the signal. Hence nuclei attached to aggregated segments of a polymer can not contribute to the NMR intensity due to lack of motional narrowing of its signal as a result of segmental arrest on aggregation. Thus a diminution of the resonance intensity as a result of a change in the solution environment has been attributed to the formation of aggregates in the solution.³

Dedicated to the memory of Professor Wunshain Fann (1961–2008).

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The solvent has manifold roles in solution NMR spectroscopy. Apart from imparting motional freedom to a solid solute, the solvent often interacts with the solute that is reflected in the NMR spectra of the latter and provides valuable information on the solution structure.

In this work we have studied the ^1H NMR spectra of a π -conjugated polymer, poly[2-methoxy-5-(2'-ethylhexyloxy)-1,4-phenylenevinylene] (MEH-PPV) in four different deuterated solvents, *viz.* toluene- d_8 ($\text{C}_6\text{D}_5\text{CD}_3$), chlorobenzene- d_5 ($\text{C}_6\text{D}_5\text{Cl}$), benzene- d_6 (C_6D_6) and chloroform- d (CDCl_3). The main focus of this work was to investigate the nature of the π - π interaction in the solution-state, which is the major force for stabilizing aggregation in π -conjugated polymer systems.⁴⁻⁶ MEH-PPV is one of the mostly studied materials of the π -conjugated polymer family. This is because of its potential use in electroluminescent display devices. The photophysics of the polymer film used in the device largely depends on the structure of the solution from which they are cast.⁷⁻¹² Thus a clear understanding of the solution structure is crucial to the fabrication of a sustainable high performance device.

RESULTS AND DISCUSSION

Fig. 1 shows the ^1H NMR spectra of 0.5% (w/v) solutions of MEH-PPV in the aliphatic solvent chloroform- d and in three aromatic solvents, benzene- d_6 , chlorobenzene- d_5 and toluene- d_8 at room temperature. In CDCl_3 the side-chain proton signals are better resolved, indicating motional narrowing of these signals due to better solvation and good mobility of the side-chain protons in this solvent. In the aromatic solvents the alkyl side-chain signals broaden and overlap, obviously due to reduced motional narrowing of these signals. This might result from solvophobic interaction of the side-chains of MEH-PPV with the aromatic solvents. However the positions of the alkyl signals do not alter much due a change from aliphatic to aromatic solvent.

In CDCl_3 the signals from alkoxy protons C and D appear very close to each other, whereas in the aromatic solvents the signals shift to lower frequencies relative to their positions in CDCl_3 and they are dispersed in the dimension of δ as a result of the interaction between these protons and the π -electron field of the aromatic solvent. This is consistent with the well known phenomenon of aromatic solvent induced shift (ASIS) of protons attached to polar groups.¹³ The observed low frequency shift of the signals indicates a non-specific interaction-geometry between the alkoxy side-groups and the aromatic solvent. This is because the

area of the shielding zone of the solvent (normal to the plane of the aromatic ring) is somewhat larger relative to the area of the deshielding zone (along the edges of the aromatic ring).¹⁴ On the contrary, however, the signals from the protons of the π -conjugated polymer backbone (A and B) suffer higher frequency shifts in aromatic solvents relative to those in CDCl_3 solution, which appear to suggest that the geometry of the interaction between the polymer backbone and the aromatic solvent favored deshielding of these protons. Now the π - π specific interaction in aromatic systems is known to favor an offset face-to-face stacked or an edge-to-face (T-shaped) geometry between the aromatic moieties.¹⁵⁻¹⁷ However, X-ray diffraction studies on solvent-cast MEH-PPV films show that the phenylene plane of the polymer stack along the crystallographic *a*-axis of the unit cell.^{18,19} Moreover, recent theoretical calculations show that the stacking geometry for π - π interaction becomes most significant for large unsaturated systems (>10 - 15 C atoms).²⁰ Hence segments of the hairy-rod type π -conjugated polymers would fall in this category. From these studies it is quite reasonable to believe that inter-segment π - π stacking might be the natural mode of interaction in π -conjugated polymer systems. The presence of such inter-

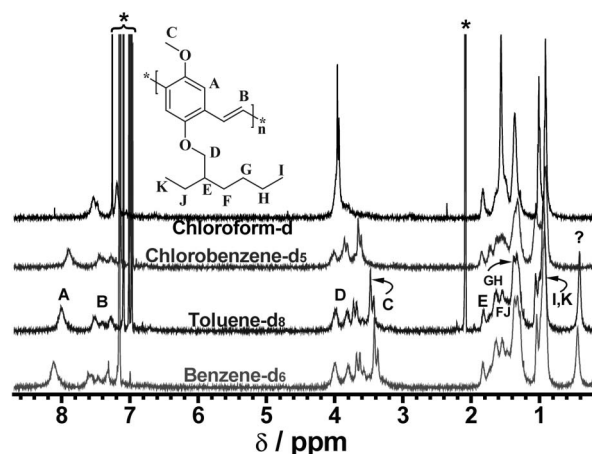


Fig. 1. 500 MHz ^1H NMR spectra of 0.5% solutions (g of polymer dissolved in 100 mL solution) of MEH-PPV in different solvents at 25 °C. The sharp lines marked with an asterisk are due to solvent residual proton resonances. δ Values are relative to TMS used as internal reference. Inset: structural formula of MEH-PPV. Non-equivalent hydrogen atoms labeled by the letters A through K correspond to their resonance bands in the spectra of the compound in toluene- d_8 solution (following the labeling of Collison *et al.*³)

Table 1. Chemical shifts (δ /ppm) of MEH-PPV protons (labeled in Fig. 1) in different solvents at 25 °C; polymer concentration 0.5% (w/v)

Solvent	A (ppm)	B (ppm)	D (ppm)	C (ppm)	E (ppm)	K ₂ I (ppm)	? (undefined) (ppm)
Chloroform-d	7.54-7.47	7.19	3.96	3.94	1.83	1.0-0.9	-
Chlorobenzene-d ₅	7.89	7.45-7.20	4.0-3.6	3.66	1.85	1.0-0.9	-
Toluene-d ₈	8.00	7.5-7.3	4.0-3.4	3.48	1.82	1.1-0.9	0.41
Benzene-d ₆	8.12	7.6-7.3	4.0-3.4	3.42	1.83	1.0-0.9	0.44

action in the solution state would be manifested by a low-frequency shift of the proton resonances of the stacked π -conjugated moieties as they would mutually lie in the shielding zone produced by the aromatic ring current.^{4,21} The appearance of the backbone proton resonances at much lower frequencies in CDCl₃ relative to their positions in the aromatic solvents (by *ca.* 0.6 ppm in benzene-d₆ for A, Table 1) indicates a stronger π - π self-stacking interaction of the polymer segments in the aliphatic solvent. The chemist's rule of thumb 'like dissolves like' predicts that an aliphatic solvent would solvate the 'hairy' aliphatic side-chains whereas an aromatic solvent would tend to solvate the conjugated backbone of the polymer. Thus a reasonable explanation for the above shift is that in CDCl₃, an unfavorable interaction between the polymer backbone and the solvent leads the MEH-PPV segments to remain close to each other forming stronger π - π complex through self-stacking. As aromatic solvents solvate the polymer backbone, they can compete for the π - π interaction with the polymer backbone and thus disrupt the polymer-polymer self-stacking tendency. Consequently the signals from the protons of the backbone π -system appear at higher frequencies in aromatic solvents relative to those in CDCl₃.⁴ As will be evident from Fig. 1, this disrupting ability increases in the order - chlorobenzene < toluene < benzene. In this context, we should mention of a literature report that suggests strong coupling between the phenyl ring of the MEH-PPV backbone and the aromatic solvent (toluene-d₈ and pyridine-d₅) through polymer-solvent π - π stacking while the vinylic protons remain unaffected.²² This causes, according to them, deshielding of the phenylic protons and a large high-frequency shift of the corresponding NMR line (A) as a consequence; while the position of the band due to the vinylic protons (B) suffers only a little shift due to lack of such interaction (i.e., assignments of the bands for A and B protons are reversed in the aliphatic solvent from those in the aromatic solvent). It should be clear from our above discussion that these authors altogether ignored the possibility

of π - π self-stacking of MEH-PPV backbone segments in aliphatic solvents, which is the principal mode of organization in the condensed phase of PPVs.^{18,19,23} As to our argument, if such stacking does occur in CDCl₃ solutions, it is more likely that the resonance of the B protons would be equally affected by changing to an aromatic solvent. As a confirmation we took NMR spectra of MEH-PPV in mixed solvent-systems containing CDCl₃ and an aromatic component over the whole composition range to look for any admixing and subsequent reversal of the signals A and B. The results of this study for the CDCl₃-C₆D₅Cl system are shown in Fig. 2. It is clearly observed that as we decrease the volume fraction of the co-solvent C₆D₅Cl, resonance bands from both the A and the B protons linearly shift to lower frequencies; there is no sign of intermediate mixing

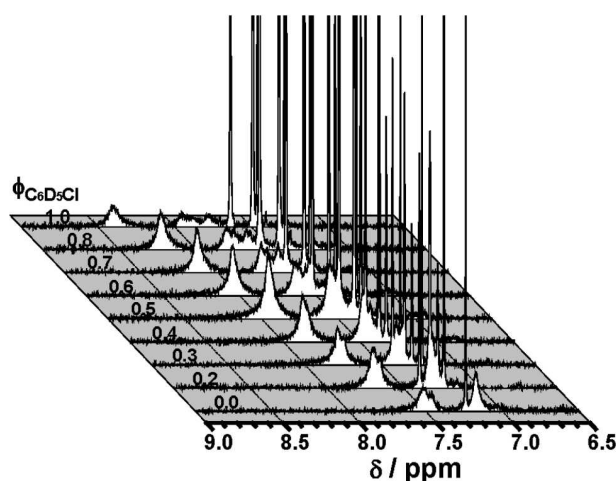


Fig. 2. 500 MHz ¹H NMR spectra of 0.5% MEH-PPV solutions in the mixed solvent system CDCl₃-C₆D₅Cl at 25 °C. The number along each spectrum indicates the volume fraction of the co-solvent C₆D₅Cl. In case of solutions in CDCl₃ and in the mixed solvents, the residual proton signal at 7.24 ppm was used as the reference. For the C₆D₅Cl solution the position of its deuterium lock signal relative to that of CDCl₃ was utilized for referencing.

and subsequent reversal of the bands. As the volume fraction of the aliphatic component in the solvent mixture is increased, polymer-solvent π - π interactions are gradually replaced by the stronger intersegment π - π stacking, and hence both of the resonance bands A and B gradually move towards lower-frequencies. Similar results have been observed for the CDCl_3 - C_6D_6 mixed-solvent system. It is therefore reasonable to retain the original assignments of the bands A and B in toluene solution due to the phenylic and vinylic backbone protons of MEH-PPV, respectively, to the corresponding bands in the aliphatic solvent, CDCl_3 .

The solutions of π -conjugated polymers are complex systems composed of a dynamic component that has been described as a transient network of polymer chains from which the solution NMR signals of the polymer are obtained and an aggregate component from which the NMR signals are lost in the baseline due to the lack of motional narrowing on segmental arrest.^{3,4,24} The aggregate component may even propagate throughout the solution on long standing leading to the formation of a gel-like jammed phase.^{25,26} A careful examination of Fig. 2 would reveal that the intensities of the MEH-PPV backbone proton resonances are higher for solutions with intermediate co-solvent compositions. Very interesting results come out as we plot the normalized integrated intensities of the polymer proton signals in the mixed solvents as a function of the solvent composition as shown in Fig. 3. The backbone aromatic proton signals are reduced to as low as 44% in chlorobenzene- d_5 -rich solvents and to 57% in chloroform- d . Such a huge loss of motional narrowing indicates a high degree of aggregation of the polymer in both the solvents. A greater loss of motional narrowing due to aggregation in chlorobenzene- d_5 rich solvents than in chloroform- d is expected because the aromatics are poorer solvent for MEH-PPV than chloroform.²² The sidechain alkyl proton signals show much higher intensity than the main chain protons and they remain higher in chloroform- d -rich solvents than the corresponding chlorobenzene- d_5 -rich compositions. This is because chloroform solvates the polymer sidechains better than chlorobenzene. Relatively lower intensity of the alkoxy proton signals than the alkyl ones may be attributed to the loss of their motional narrowing as a result of the nearness of their attachment to the aggregating chain backbone. However, the signal intensities gradually increase and pass through a maximum in the intermediate co-solvent compositions. This behavior may be related to the difference in the nature of the aggregation of the MEH-PPV

chains in aromatic and aliphatic solvents. In the aromatic solvent, the solvent molecules can compete with the polymer against polymer self-aggregation and the former being the predominant species, aggregation of the polymer-solvent π - π complex is favorable. On the other hand in chloroform- d , due to the absence of the polymer backbone-solvent interaction, polymer-polymer π - π stacking can lead to the aggregation. As we increase the chloroform- d fraction in the solvent mixture, it pulls apart the polymer chains of the prevalent polymer-solvent aggregates by solvating the side-chains of the polymer thus increasing motional narrowing. At higher volume fractions of the co-solvent CDCl_3 , the abundant aliphatic solvent species replaces the aromatic solvent. As CDCl_3 can not solvate the backbone π -system, desolvation occurs that leads to the formation of aggregates through polymer-polymer π - π stacking and consequent loss of motional narrowing of the NMR signals.

There is a clear signal centered at 0.44 ppm in benzene- d_6 and at 0.41 ppm in toluene- d_8 solutions, which are absent from the spectra of chlorobenzene- d_5 and chloroform- d solutions (Fig. 1). In previous works of Collison *et al.*³ and Traiphol *et al.*²² this prominent signal has been left

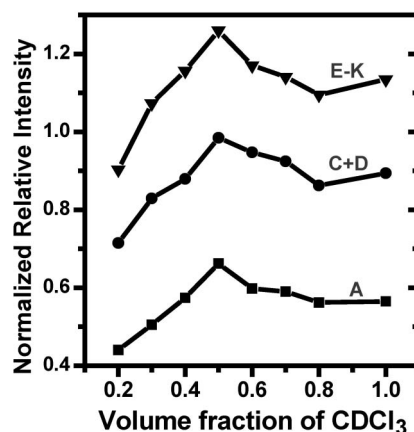


Fig. 3. Intensities of MEH-PPV protons normalized to the intensity of the residual ^1H signal of CDCl_3 (0.2 atom% H) as a function of the composition of the mixed-solvent system CDCl_3 - $\text{C}_6\text{D}_5\text{Cl}$ at 25 °C corresponding to the spectra shown partly in Fig. 2. The labeling of the protons is shown in Fig. 1. [A value greater than 1 for the normalized intensity might have contributions from: (a) the uncertainty of proton integration ($\sim 5\%$), (b) uncertainty in the level of deuteration of the solvent and (c) uncertainty in the concentration of the solutions as these were prepared on a very small scale].

unnoticed (the former group did not show any spectrum in their paper while the latter group did not show the 0–0.5 ppm part of the spectrum). This undefined band is not due to any impurity in the sample as it does not appear in Chloroform-*d* or Chlorobenzene-*d*₅ solutions. The line is not due to the terminal methyl groups of the side chain (6H) either, as it corresponds only to 1.6 protons in toluene-*d*₈ and to 1.3 protons in benzene-*d*₆ relative to the intensity of the lines between 0.2 and 2 ppm (due 15 H in the side-chain of MEH-PPV). Also it can not be due to water because the band appears in a thoroughly dried sample (by bubbling dry nitrogen through the solution for *ca.* 1 hour before taking the spectra) as well. Moreover, we have verified that in toluene-*d*₈ solution, its position is relatively insensitive to the temperature of the measurement. At sub-ambient temperatures, however, the intensity of the band gradually reduces and it becomes indistinguishable with the baseline below *ca.* –20 °C (there are other important features of the low-temperature spectra that are under investigation and would appear elsewhere). Toluene (also benzene) is a poor-solvent for the side-chain, which can lead to the aggregation of some of the side-chains and hence the protons contained in them may give rise to the broad resonance band due to slower dynamics of the aggregated side-chains (Fig. 4). The diminution of the band at sub-ambient temperatures may also indicate its association with some aggregated structure that freezes when the temperature is reduced.

EXPERIMENTAL SECTION

Poly[2-methoxy-5-(2'-ethylhexyloxy)-1,4-phenylene

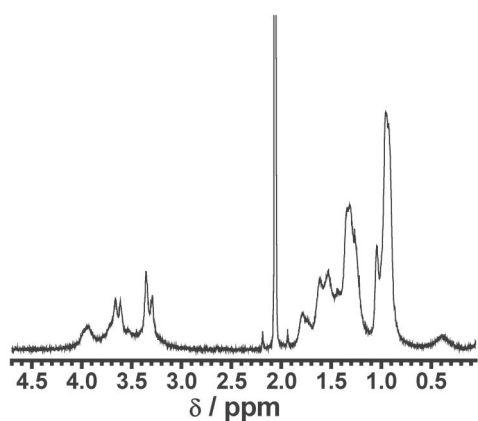


Fig. 4. 500 MHz ¹H NMR spectra of a 0.5% solution of MEH-PPV in C₆D₅CD₃ at –20 °C. The band at 0.4 ppm is much broader and its intensity has been significantly diminished relative to that of the room temperature spectra (*vide* Fig. 1).

vinylene] (MEH-PPV) was obtained from Sigma-Aldrich corporation. Its weight averaged molecular weight (*M_w*) and polydispersity index (PDI) were 18500 and 3.1, respectively, as determined by gel-permeation chromatography (GPC) using tetrahydrofuran (THF) as the eluting solvent against polystyrene standards. Deuterated solvents benzene-*d*₆ (99.6 atom% D), chlorobenzene-*d*₅ (99 atom% D), toluene-*d*₈ (99.6 atom% D) and chloroform-*d* (99.8 atom% D) were products of Cambridge Isotope Laboratories, Inc., U.S.A.

Weighed amount of the MEH-PPV was dissolved in 0.5 mL of the solvent by stirring 72 hours at room temperature (*ca.* 25 °C) prior to NMR measurements. The toluene solution had visual gel-like particles dispersed and was filtered through 5 μm TEFLON filter. Other solutions were clear to the eye. The final concentration of the solution was determined from the height of the solution in the NMR tube (5 mm outer diameter) calibrated to the volume. ¹H NMR measurements were conducted at 499.84 MHz using a Varian UnityInova-500 NMR spectrometer at the department of Chemistry of the National Tsing-Hua University, Hsinchu, Taiwan. For comparison purpose, the peak at 0.0 ppm from TMS used as internal standard was the reference. Where appropriate, the residual solvent signal centered at 2.09 ppm of toluene-*d*₈ and that at 7.24 ppm of chloroform-*d* were also utilized for referencing purpose. For solvent mixtures with chloroform-*d*, the 7.24 ppm signal was the reference. For solutions with benzene-*d*₆ and chlorobenzene-*d*₅ the position of their deuterium lock signal relative to that of chloroform-*d* were utilized for referencing.

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